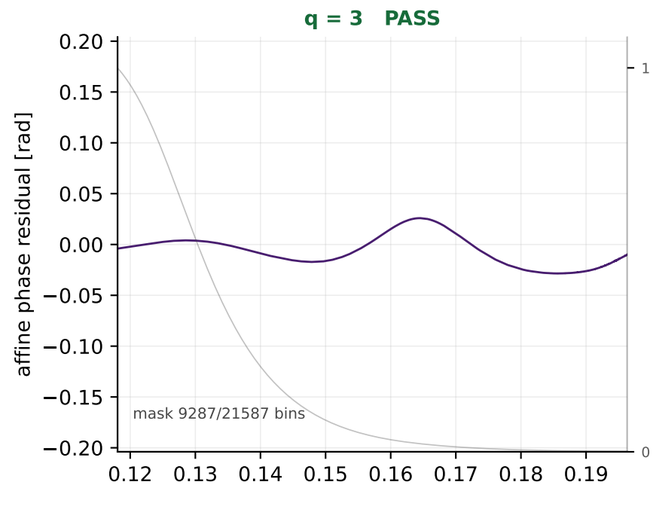
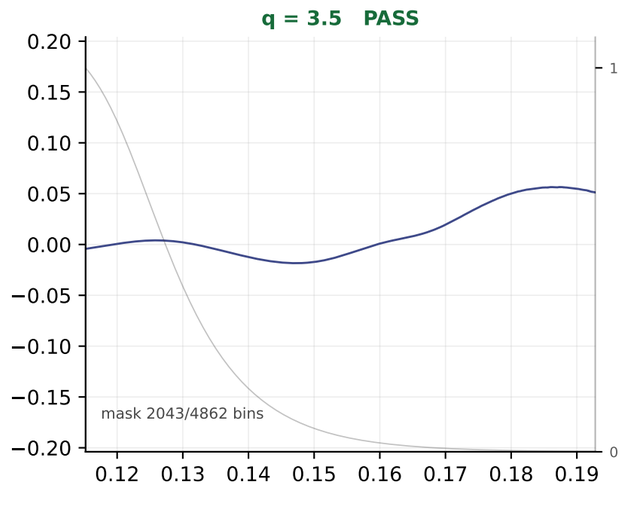
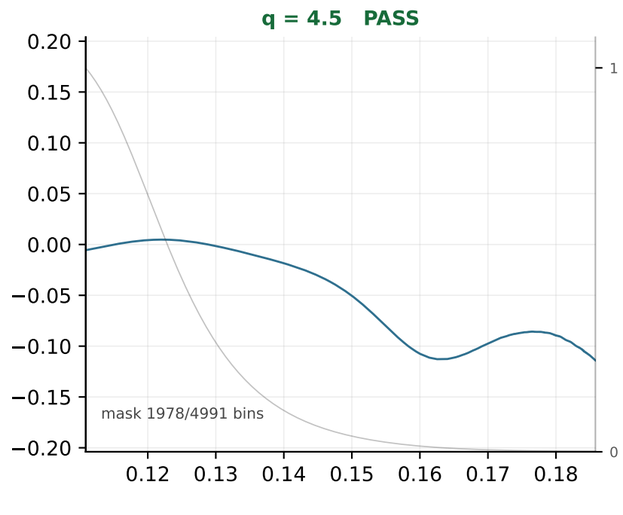
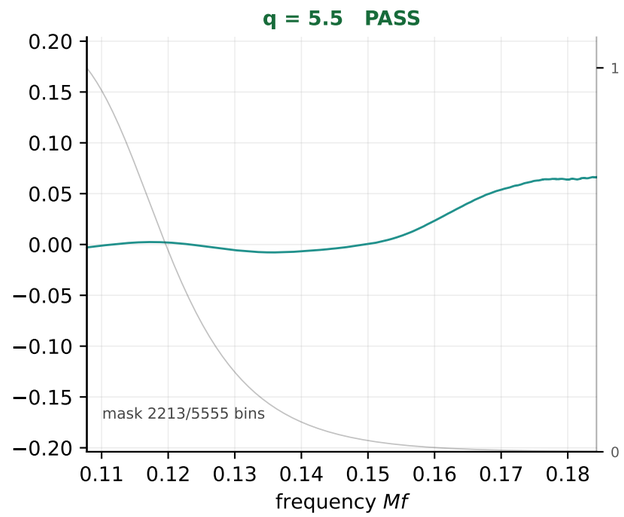
NR33 phase benchmark: frozen ringdown-only single-mode comparison

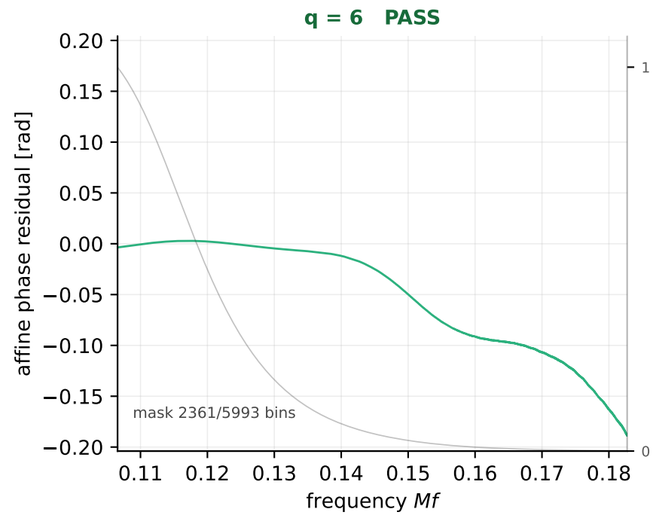
NR reference vs public LAL PhenomXHM (release 122022), 60 Msun phase curves; curves end at the phase mask; gray line = normalized NR amplitude

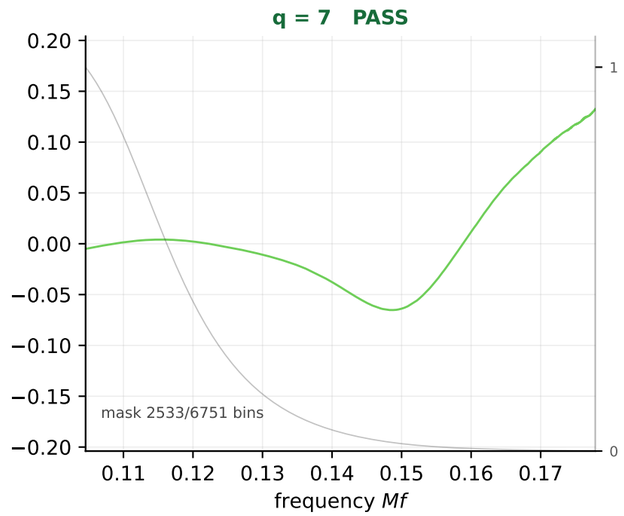


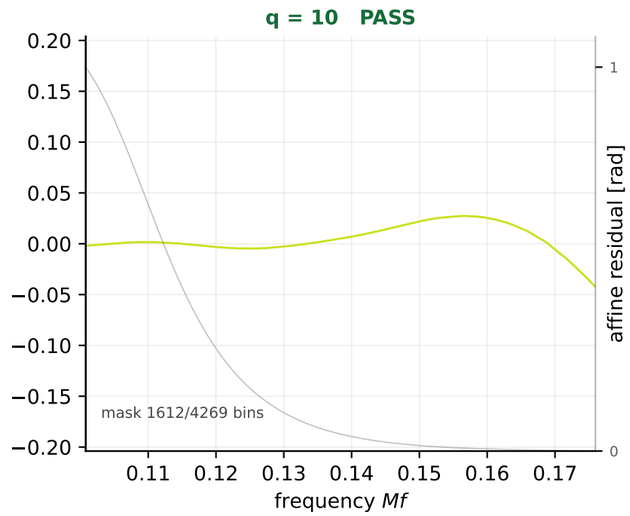


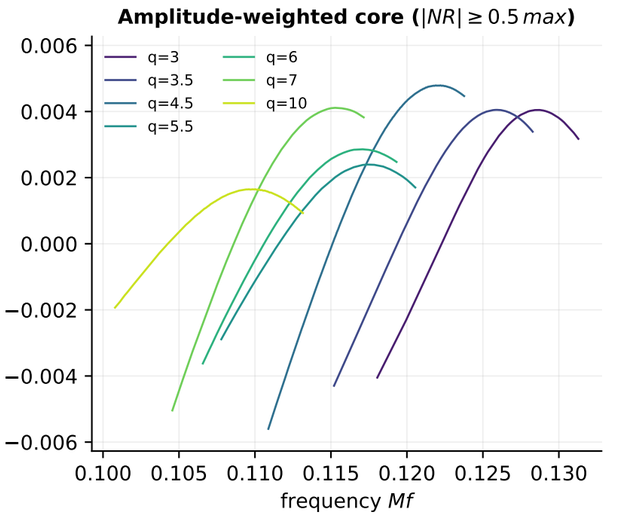


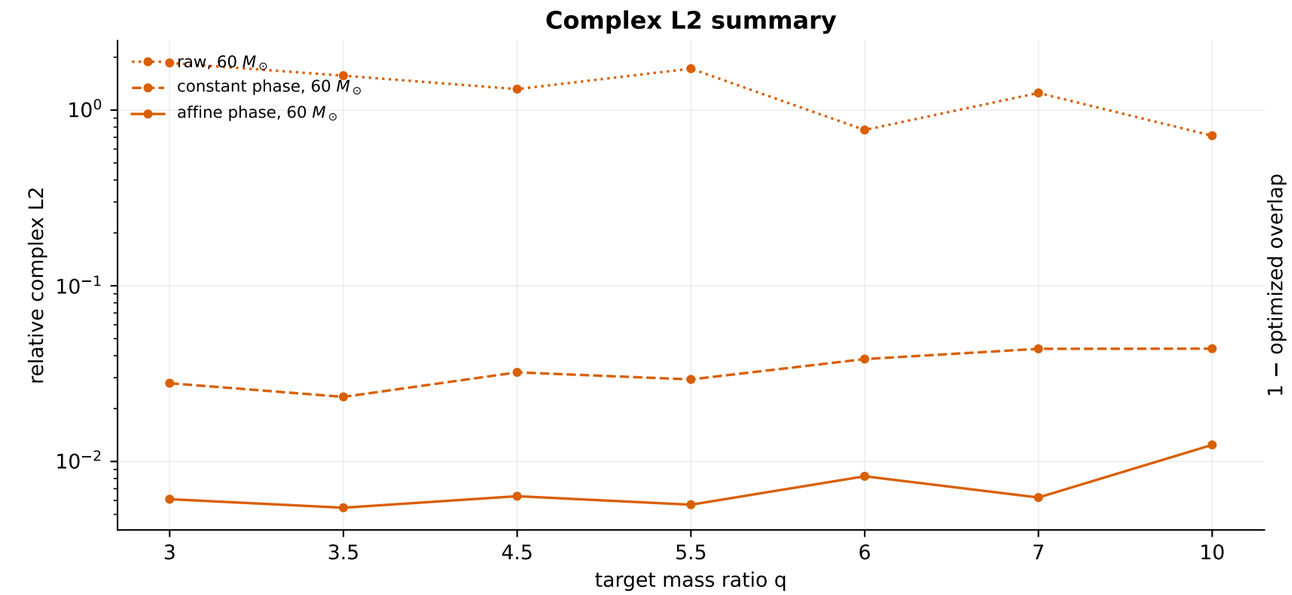














Affine phase fit is NR-amplitude-squared weighted on the preregistered mask. Summary metrics are read from the frozen report for 20, 60, and 120 Msun.